

Azure Cloud Hands-on Project

Problem Statement: Smart Document Management System

Background

A company currently stores employee and business documents on individual computers, making it difficult to manage, search, and retrieve files. They want to migrate to Azure Cloud by building a secure document management API that stores files in Azure Blob Storage while maintaining metadata in Azure SQL Database.

Your task is to build a **FastAPI** application that integrates with **Azure Blob Storage** and **Azure SQL Database**.

Objective

Develop a FastAPI application that enables users to:

- Upload documents
- Store files in **Azure Blob Storage**
- Store document metadata in **Azure SQL Database**
- View all uploaded documents
- View document details by ID
- Download documents
- Delete documents
- Generate **secure, time-limited sharing links** that automatically expire after a specified duration (e.g., 5 minutes, 1 hour, or 24 hours).

The application may be run **locally**. Deploying it to **Azure App Service** is optional.

Azure Services to be Used

Mandatory

- Azure Subscription
- Resource Group

- Azure Storage Account (Blob Storage)
- Azure SQL Database

Optional

- Azure App Service
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Functional Requirements

1. Upload Document

The application should allow users to upload files (PDF, Image, or Word document).

For every uploaded file:

- Store the file in Azure Blob Storage.
 - Store the metadata in Azure SQL Database.
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2. View All Documents

Create an API that returns a list of all uploaded documents.

3. View Document by ID

Create an API to retrieve the metadata of a specific document.

4. Download Document

Allow users to download the document stored in Azure Blob Storage.

5. Generate Secure Share Link

Provide an API that generates a **time-limited secure URL (SAS URL)** for a document.

Requirements:

- User specifies the expiry duration (for example, 5 minutes, 1 hour, or 24 hours).
 - The generated link should automatically become invalid after the expiry time.
 - Only users possessing the generated link should be able to access the document while it is valid.
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6. Delete Document

When a document is deleted:

- Remove the file from Azure Blob Storage.
 - Remove the corresponding metadata from Azure SQL Database.
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Database Schema

Documents

Column	Type
document_id	INT
file_name	VARCHAR
blob_url	VARCHAR
uploaded_by	VARCHAR
upload_date	DATETIME
file_type	VARCHAR

Azure Infrastructure Requirements

Students should create:

- Azure Subscription (provided)
- Resource Group
- Storage Account
- Blob Container
- Azure SQL Database

Configuration

Store secrets using environment variables.

Example:

- `AZURE_STORAGE_CONNECTION_STRING`
 - `AZURE_CONTAINER_NAME`
 - `AZURE_SQL_CONNECTION_STRING`
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Deliverables

- GitHub Repository containing the complete source code
 - README with setup instructions and sample API requests/responses
 - Screenshots demonstrating successful upload, document listing, secure share link generation, and deletion
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Evaluation Criteria

1. Azure Integration (50%)

Correct implementation of Azure Blob Storage and Azure SQL Database with proper resource organization.

2. Application Functionality (50%)

FastAPI application successfully performs document upload, retrieval, secure time-limited sharing, and deletion, with the source code maintained in a GitHub repository.